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**CLASSIFICATION OF HYPERSPECTRAL IMAGES USING
MACHINE LEARNING AND DEEP LEARNING METHODS**

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SENIOR PROJECT

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LIST OF ABBREVIATIONS

1D-CNN	1-Dimensional Convolutional Neural Network
3D-CNN	3-Dimensional Convolutional Neural Network
AE	Autoencoder
R&D	Research and Development
CNN	Convolutional Neural Networks
GAN	Generative Adversarial Networks
GPU	Graphics Processing Unit
GUI	Graphical User Interface
HSI	Hyperspectral Images
MRF	Markov Random Fields
PCA	Principal Component Analysis
RF	Random Forest
RNN	Recurrent Neural Networks
SAE	Stacked Autoencoders
SVM	Support Vector Machines
ViT	Vision Transformer

CLASSIFICATION OF HYPERSPECTRAL IMAGES USING MACHINE LEARNING AND DEEP LEARNING METHODS

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Department of Computer Engineering
Senior Project

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This project develops a unified classification framework for hyperspectral images (HSI) that spans supervised, semi-supervised, and federated learning paradigms, with the overarching goal of characterizing how each approach handles the dual challenge of high spectral dimensionality and scarce labeled samples in remote sensing contexts. Three deep learning architectures form the core of the system: 2D-CNN, 3D-CNN, and Vision Transformer (ViT). Raw spectral data exceeding 200 bands is reduced to 30 principal components through PCA, preceded by Gaussian filtering for noise suppression and channel-wise normalization. Semi-supervised training is realized via a self-training strategy augmented by curriculum learning, which structures the presentation of unlabeled samples from confident to uncertain predictions to stabilize optimization. Privacy-preserving collaborative training is achieved through a simulated server-client federated architecture operating over three standard benchmark datasets: Indian Pines, Pavia University, and Salinas. All models are assessed on Overall Accuracy, Precision, Recall, F1-Score, and Cohen's Kappa coefficient. The results reveal a clear performance split. Semi-supervised learning reaches competitive accuracy on the two larger datasets (86–97%) but falls substantially short on the smaller Indian Pines set (54–74%), yielding an overall average below supervised baselines. Federated learning proves more resilient across datasets, incurring an average accuracy drop of only 3.1% relative to centralized training while preserving data privacy. A PyQt6-based GUI with multithreaded inference makes the trained models accessible for real-time HSI classification without

requiring code-level interaction.

Keywords: Hyperspectral Image Classification, Deep Learning, CNN, Vision Transformer, Semi-Supervised Learning, Federated Learning, Machine Learning, Curriculum Learning, Remote Sensing, Principal Component Analysis

HİPERSPEKTRAL GÖRÜNTÜLERİN MAKİNE ÖĞRENMESİ VE DERİN ÖĞRENME YÖNTEMLERİYLE SINIFLANDIRILMASI

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Bitirme Projesi

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Bu proje, hiperspektral görüntülerin (HSI) sınıflandırılması için denetimli, yarı-denetimli ve federe öğrenme paradigmalarını bir arada ele alan bütünlük bir çerçeve sunmaktadır. Temel motivasyon, uzaktan algılama alanında sıklıkla karşılaşılan iki temel güçlükten kaynaklanmaktadır: yüksek spektral boyutsallık ve yetersiz etiketli örnek sayısı. Sistemin omurgasını üç derin öğrenme mimarisi oluşturmaktadır: 2D-CNN, 3D-CNN ve Vision Transformer (ViT). 200'ü aşkın spektral banttan oluşan ham veri, kanal bazlı normalizasyon ve Gaussian gürültü filtrelemesinin ardından Temel Bileşen Analizi (PCA) ile 30 bileşene indirgenmektedir. Yarı-denetimli eğitimde, müfredat öğrenmesiyle güçlendirilmiş bir kendinden eğitim stratejisi benimsenmiştir; bu yaklaşım, etiketsiz örnekleri güven skoruna göre kolaydan zora sıralayarak optimizasyon sürecini kararlı kılmaktadır. Federe öğrenme ise üç standart kıyaslama veri seti (Indian Pines, Pavia University ve Salinas) üzerinde çalışan simüle edilmiş bir sunucu-istemci mimarisi aracılığıyla hayata geçirilmiş, gizlilik korumalı dağıtık eğitime olanak tanımıştır. Modeller; Genel Doğruluk, Kesinlik, Geri Çağırma, F1-Puanı ve Cohen's Kappa katsayısı metrikleriyle karşılaştırmalı biçimde değerlendirilmiştir. Sonuçlar belirgin bir ayrışma ortaya koymaktadır. Yarı-denetimli öğrenme, iki büyük veri setinde rekabetçi doğruluk elde ederken (%86–97), daha küçük Indian Pines veri setinde denetimli temel çizgisinin önemli ölçüde gerisinde kalmaktadır (%54–74). Federe öğrenme ise veri setleri genelinde daha tutarlı bir performans sergileyerek merkezi eğitime kıyasla

yalnızca ortalama %3,1 doğruluk kaybıyla gizlilik güvencesi sunmaktadır. Geliştirilen PyQt6 tabanlı, çok iş parçacıklı GUI, eğitilmiş modelleri kod düzeyinde müdahale gerektirmeksizin gerçek zamanlı HSI sınıflandırması için erişilebilir kılmaktadır.

Anahtar Kelimeler: Hiperspektral Görüntü Sınıflandırması, Derin Öğrenme, CNN, Vision Transformer, Yarı-Denetimli Öğrenme, Federe Öğrenme, Makine Öğrenmesi, Müfredat Öğrenmesi, Uzaktan Algılama, Temel Bileşen Analizi

1

INTRODUCTION

With the ability to capture electromagnetic signals across many narrow spectral bands, Hyperspectral imaging (HSI) reflects electromagnetic signatures across hundreds of spectral bands, revealing properties of the physical and chemical structures of the surface of the earth. Precise details at this level are not possible with conventional RGB imagery. HSI's capability to capture detailed spectral information make it highly useful in many fields including agriculture, environmental monitoring and defense. However, the advanced spectral information comes with the fundamental computational burden of the curse of dimensionality. When considering the scarcity of expertly annotated pixel samples, the cost of training machine learning and deep learning models increases significantly along with the model's generalizability.

The gap in the performance of supervised versus semi-supervised deep learning (CNN, ViT, Self-Training) architectures due to high dimensionality and the scarcity of labeled data is the main focus of this research work. Two primary things set this work apart. One, a curriculum learning approach is used to the training of these models in a bid to stabilize optimization. Two, a multi-threaded GUI was created to the point where users can load their own data, get predictions, and do all this without needing to code or even touch the command line.

The remaining chapters of this report review and analyze our work in these areas. Chapter 2 focuses on analyzing the state of the art research. Design and methodology of the entire system including data processing and architecture are discussed in Chapter 3. Chapter 4 provides a set of results based on quantitative comparative experiments and Chapter 5 provides analysis of these results and proposes ideas and recommendations for future enhancements.

1.1 Purpose

The primary aim of this project is to describe, in detail, the performance disparity between supervised and semi-supervised learning as applied to the classification of

high-dimensional hyperspectral data. The aim of this project can be broken into the following sub-aims:

- To evaluate the performance of supervised and semi-supervised models on the standard hyperspectral datasets: Indian Pines, Pavia University, and Salinas.
- To evaluate the impact of Gaussian filtering and PCA on the classification of hyperspectral data.
- To introduce Curriculum Learning in training to each model to improve training stability and convergence.
- To carry out a balanced and fair assessment of each model.
- To develop a Python multithreaded GUI, allowing end users to use the models to classify data in real time and with no need to access the code.

1.2 Preliminary Investigation

There is an observable shift in pattern from using traditional machine learning models towards using models built on deep learning architectures in the literature on hyperspectral image classification. Current models face problems when dealing with big data since most are trained using a lot of labeled data. Constructing labeled data for hyperspectral images is an expensive and time-consuming endeavor due to the need for an expert in the field to assist with the data labeling.

This project aims to fill important gaps in the literature. While most models translate well to theory, they end up being just a concept, since most of them are scripts that require technical expertise to use. The application, based on the experimental code, uses a that gives users the ability to input their own datasets and receive results in near real time.

2 LITERATURE ANALYSIS

Hyperspectral image (HSI) classification has been a focus area of remote sensing for many years, especially for being able to identify surface materials, something that broadband sensors are unable to do. There are challenges in the field that require a balance between deep models and the limited amount of training data that has been labeled. This section puts HSI classification methods into groups based on the models used and the challenges that they solve in order to help understand the development of these methods.

2.1 Traditional Machine Learning Approaches

In the beginning, HSI classification used shallow and statistical methods of machine learning. Dominant frameworks in this time were Support Vector Machines (SVM) and Random Forest (RF). An example of early work in HSI image classification was by Du et al. in using sparse discriminative representations and multinomial logistic regression to achieve 97.71

Although the methods mentioned previously are robust to noise and interpret output, they have become limited in terms of classification accuracy. A challenge for machine learning is to capture the associations in high-dimensional spectral spaces. Classical techniques in feature extraction also struggle to meet the accuracy of representations that deep networks learn.

2.2 Convolutional Neural Network-Based Approaches

Learning spatial and spectral features simultaneously and without manual feature engineering has changed HSI classification with CNNs. 3D-CNNs with L2 regularization and dropout were applied by Chen et al. and achieved a groundbreaking 99.66

A challenge of 3D-CNNs is the parameter cost. Therefore, some strategies were introduced. AcousticSN was able to balance spectral-spatial expressiveness against computational cost by combining both 2D and 3D convolutions. In a different approach, Ahmad et al. introduced small 3D-CNN architectures with faster inference times that maintained the same accuracy. Boundary artifacts were addressed by Paoletti et al. by mirroring borders, which mitigated information loss for the edges of each patch. Combined, these approaches provided a strong justification for deep learning features with CNNs for HSI classification, but their requirements for labeled data and GPU memory grow with depth, which limits their use in real-world situations with a lack of data.

2.3 Autoencoders, Generative Networks, and RNNs

Data that is insufficiently labeled has prompted the use of architectures that give representations of a decent quality without much supervision. For instance, Stacked Autoencoders (SAE) offer a good method of performing unsupervised feature extraction by dimensionality reduction. Combining SAEs with CNNs, as done by Li et al., focuses on learning spatial weights that are adaptive to one locality, while Zhao et al. focused on SAE-based compression and 3D deep residual networks that combine both dimensionality reduction and classification, but assume a significant amount of labeled data.

Sequence modeling in HSI began with Recurrent Neural Networks. In their work, Shi et al. employed RNNs to incorporate spectral-spatial features of different scales. They considered different spectral bands to be in a temporal sequence. Another approach to the problem was Generative Adversarial Networks, which synthesized samples and improved small labeled datasets and, in turn, aimed to mitigate the generalization issue caused by the lack of labeled samples.

Each of these frameworks imposes additional issues. SAE and RNN frameworks suffer from the burden of sequential computation, and update the systems slowly and GANs offer unpredictable results. Small changes to hyperparameters can annihilate generators or result in mode collapse and create uninformative training data.

2.4 Transformer Architectures and Semi-Supervised Methods

In the realm of HSI classification, Transformers improve on capturing long range spectral-spatial relations. Their addition to CNN based local feature extraction has also produced positive results. Recent tokenization methods referring to the encoding

of spectral-spatial areas as sequences have allowed more unmediated use of the Vision Transformer. Thanks to recent architectures featuring dual attention, benchmark results on various datasets have surpassed 99

Transformers have proven especially powerful in this field, but their increased architecture cost is not overshadowed by previous methods. Their cost and hardware demands are greater for large datasets and decrease their usefulness in low-data scenarios.

In the realm of semi-supervised learning, Sellami et al. showed that 3D deep networks, combined with adaptive band selection, leverage unlabeled data effectively and reduce the gap in accuracy of the labeled and unlabeled data. This research serves as an excellent proof of concept. Most of the problems this research identifies, like decision boundary bias regarding pseudo-label noise and confidence threshold selection, are still unresolved in the domain.

2.5 Current Standing and Original Contribution of This Study

A common theme in the reviewed research is that the models that are the most accurate when a lot of labeled data are available are also the most fragile models when labeled data are sparse, and vice versa. Transformers and deep CNNs tend to overfit when trained on small labeled datasets. Semi-supervised and generative approaches to the data problem introduced instability to the training process and sensitivity to thresholds, and no previous research has incorporated stability directly into the semi-supervised training process.

This research aims to solve this issue by combining CNNs and Transformers in a Self-Training framework incorporating Curriculum Learning. This is the technique that organizes the training samples in order of the estimated level of difficulty so that the model does not focus on the noisy pseudo-labels early on in the training process. In addition to its novelty, most of the other research in the domain consists of models that only function through command-line interfaces, which makes them inaccessible to most practitioners that do not have a specialized knowledge of deep learning. The work presented here aims to convert research models into applications that can be deployed and used outside of an academic environment through the introduction of a multithreaded PyQt6 GUI. It also aims to provide real-time predictions using other external datasets.

System analysis helps bridge design and problem identification. It tells us what must be done, the limitations and constraints the system will be bound within, and the requirements for measuring the quality of the outputs the system produces. In this chapter, project goals are described, the methods and processes of requirements elicitation are detailed, and the proposed solution is examined for feasibility in terms of technology, law, economy, and labor.

3.1 System Analysis

As this is an RD project, requirements were not provided because there were no client specifications. Instead, requirements were developed through the analysis of issues and constraints provided by existing deep learning based HSI classifier algorithms. The main function of the system is to read and handle high-dimensional spectral data (either introduced externally by the user or extracted from established benchmarks - Indian Pines, Pavia University, and Salinas) and classify high accuracy pixel data.

The system is governed by three functional requirements:

- Raw hyperspectral cubes must be filtered using a Gaussian filter and PCA [1].
- Supervised (CNN [2], ViT [3]) and semi-supervised (Self-Training [4]) models must be trained in a Curriculum Learning framework [5] that regulates the difficulty of samples.
- A multithreaded GUI must be used to encapsulate all trained models, allowing users to perform simultaneous model testing and view per pixel predictions in real-time without modifying the underlying source code.

Meeting these requirements will be evaluated using Accuracy, Precision, Recall, F1-Score, and Cohen's Kappa coefficient.

3.2 Feasibility

The analysis examines the likelihood of the proposed solution in four categories: technical, legal, economic, and workforce.

3.2.1 Technical Feasibility

The proposed programming language is Python [6] since it has a complete ecosystem for machine learning with a favorable community. Deep learning is constructed with PyTorch [7] or TensorFlow [8] with data management offered by Scikit-learn [1] and OpenCV [9]. The design of the graphical user interface is done with PyQt6. The choice of the PyQt6 library was made because of its built-in support for multithreaded worker patterns, which is important for keeping the user interface responsive when time-consuming inference operations are performed. The construction is done with VS Code [10] and the versioning is done with GitHub. [11].

3.2.1.1 Hardware Feasibility

Training deep models with long inference operations on complete hyperspectral cubes requires high computational resources. Google Colab [12], provides the needed graphic processing units and eliminates the need for the computational resources for the project. The graphic user interface and integration design are done on the project team's personal macOS systems.

3.2.2 Legal Feasibility

All libraries in the stack of PyQt6, Python, PyTorch, and OpenCV are fully open-sourced. The three benchmark datasets in use are legally permitted for academic use. There are no licensing or patent barriers.

3.2.3 Economic Feasibility

The entire tool chain is free and open-source with Google Colab offering the graphic processing units at no cost. The project makes no additional capital investments with the teams' personal systems.

3.2.4 Workforce and Time Feasibility

The project team of Tan Erciyas and Kerem Baydar has the required skills to execute the entire project.

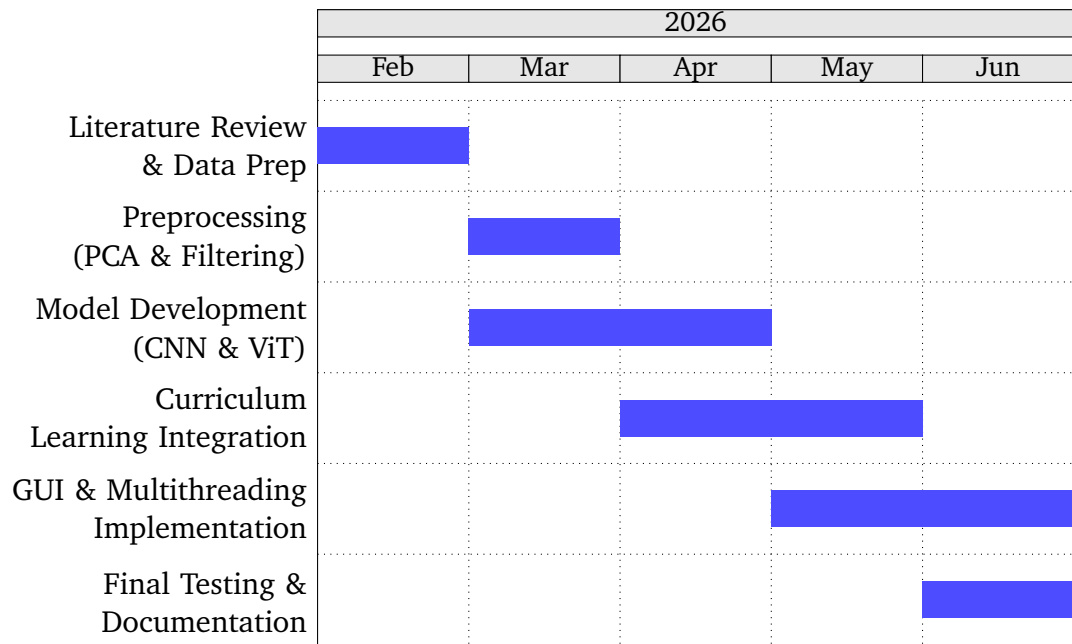


Figure 3.1 Project Gantt Chart (Feb 2026 - June 2026)

4

SYSTEM DESIGN

Translating requirements into an architectural framework occurs in system design. This chapter describes the datasets, preprocessing pipelines for spectral data, the three system core deep learning architectures, and the supervised, semi-supervised, and federated training methods examined.

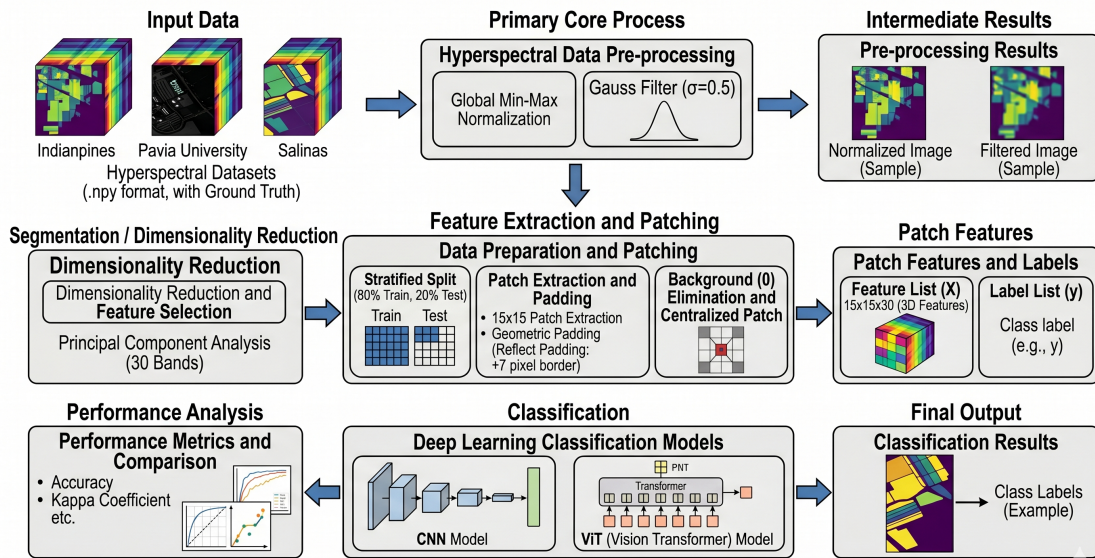


Figure 4.1 Overall Workflow of the Proposed HSI Classification System

4.1 Material and Dataset

The performance of different models on a classification task in hyperspectral imaging (HSI) is highly dependent on the different features of the used dataset. These features include the model's spatial resolution, the level of class imbalance, and the number of samples in the dataset with labels. In this work, we use three benchmark datasets from the remote sensing literature, and each of these datasets has its own set of challenges.

4.1.1 Dataset Specifications

4.1.1.1 Indian Pines

This dataset is collected by the AVIRIS sensor and has a 145x145 grid with 224 spectral bands on 16 vegetation and land-cover classes. In most cases, the water absorption bands reduce the number of usable bands to 200.

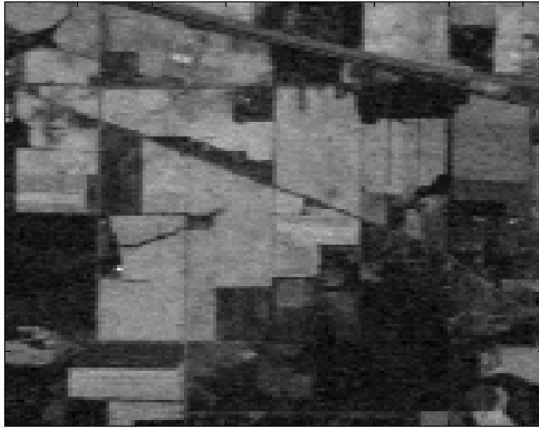


Figure 4.2 Indian Pines – False-color band composite

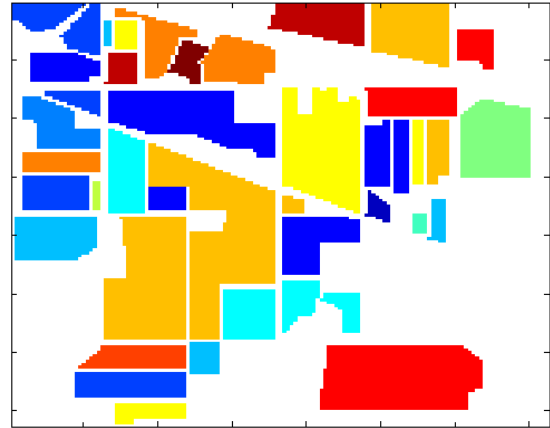


Figure 4.3 Indian Pines – Ground truth map

Table 4.1 Groundtruth classes for the Indian Pines scene and their respective samples number

#	Class	Samples
1	Alfalfa	46
2	Corn-notill	1428
3	Corn-mintill	830
4	Corn	237
5	Grass-pasture	483
6	Grass-trees	730
7	Grass-pasture-mowed	28
8	Hay-windrowed	478
9	Oats	20
10	Soybean-notill	972
11	Soybean-mintill	2455
12	Soybean-clean	593
13	Wheat	205
14	Woods	1265
15	Buildings-Grass-Trees-Drives	386
16	Stone-Steel-Towers	93
Total		10249

4.1.1.2 Pavia University

This dataset is collected by the ROSIS sensor and covers 610x340 pixels with 103 spectral bands. This dataset has 9 urban land-cover classes which makes this dataset a very challenging multi-class urban benchmark.



Figure 4.4 Pavia University – False-color band composite

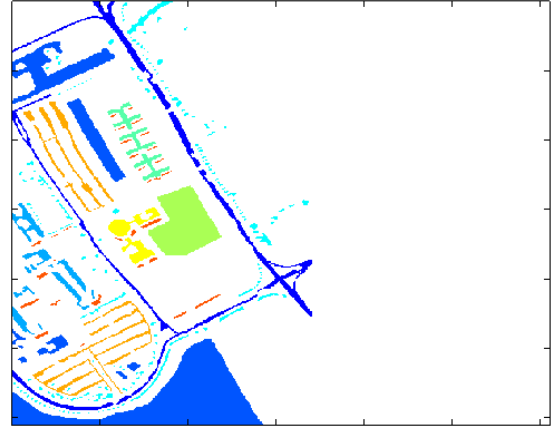


Figure 4.5 Pavia University – Ground truth map

Table 4.2 Groundtruth classes for the Pavia University scene and their respective samples number

#	Class	Samples
1	Asphalt	6631
2	Meadows	18649
3	Gravel	2099
4	Trees	3064
5	Painted metal sheets	1345
6	Bare Soil	5029
7	Bitumen	1330
8	Self-Blocking Bricks	3682
9	Shadows	947
Total		42776

4.1.1.3 Salinas Scene

This dataset is also collected by the AVIRIS sensor but this time over the Salinas Valley of California. This dataset has a spatial resolution of 512x217 pixels which is also the highest resolution of the other two datasets. This dataset also contains 224 spectral bands which are reduced to 204 after removal of the water absorption bands.

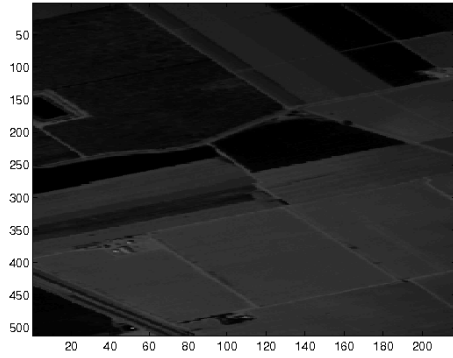


Figure 4.6 Salinas Scene – False-color band composite

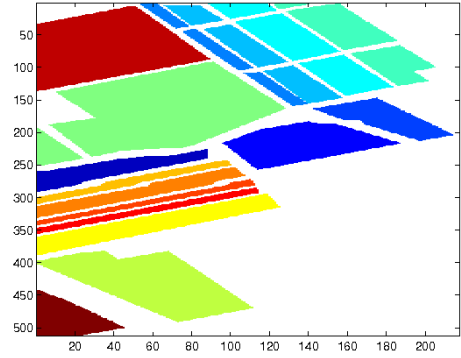


Figure 4.7 Salinas Scene – Ground truth map

Table 4.3 Groundtruth classes for the Salinas scene and their respective samples number

#	Class	Samples
1	Brocoli_green_weeds_1	2009
2	Brocoli_green_weeds_2	3726
3	Fallow	1976
4	Fallow_rough_plow	1394
5	Fallow_smooth	2678
6	Stubble	3959
7	Celery	3579
8	Grapes_untrained	11271
9	Soil_vinyard_develop	6203
10	Corn_senesced_green_weeds	3278
11	Lettuce_romaine_4wk	1068
12	Lettuce_romaine_5wk	1927
13	Lettuce_romaine_6wk	916
14	Lettuce_romaine_7wk	1070
15	Vinyard_untrained	7268
16	Vinyard_vertical_trellis	1807
Total		54129

4.1.2 Data Structure and Storage

All of these datasets have the same 3D structure and can be visualized as a hypercube, where the dimensions of the hypercube are the spatial dimensions (H and W) and the number of spectral bands (B). All datasets are stored as NumPy arrays (.npz) and are loaded in spatial patches to facilitate the training process. This supports the local spatial context and allows us to use the limited memory efficiently.

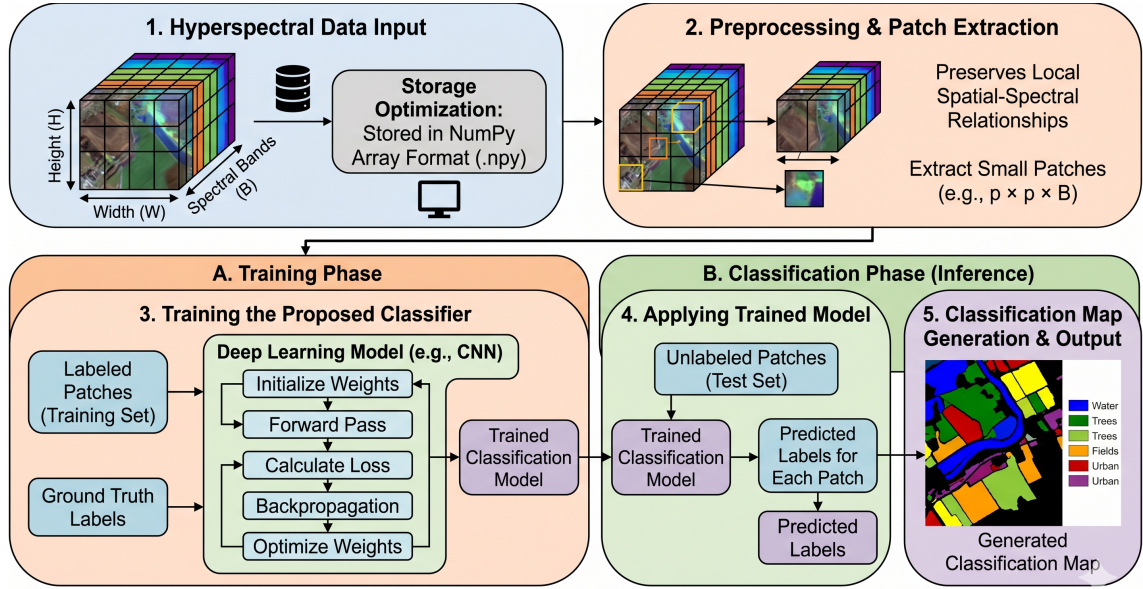


Figure 4.8 General Block Diagram of the Proposed HSI Classification System

4.2 Methodology

The processing pipeline consists of three sequential phases: preprocessing, patch-based feature extraction, and learning. Each phase is connected to the next, and the characteristics of hyperspectral data are reflected in the design of each phase.

4.2.1 Preprocessing and Dimensionality Reduction

Hyperspectral cubes are made of hundreds of noises and correlated bands and can mislead learned features. Consequently, the cost of training can increase greatly. In order to mitigate this, three operations are done.

- 1. Global Min-Max Normalization:** In this step, the hyperspectral data is reshaped from a 3D hypercube to a 2D matrix. From here, the global minimum and maximum across all pixels and bands of the reshaped data are calculated. Each value is then rescaled per the new global minimum and maximum.

Normalization of the bands was avoided to prevent variance loss.

$$\hat{x} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad (4.1)$$

2. **Spatial Gaussian Filtering:** In this step the data is filtered using a Gaussian Filter. The value of σ was set to 0.5 and was only applied to the spatial axes of the 3D hypercube. A value of zero was set for the spectral σ ensuring that the relationships between the bands of the data were preserved.
3. **Fixed-Component PCA:** In this step, PCA is applied with a value for $n_components$ set to 30. Prior to this, a variance threshold of (99.5%) was applied. However, for both the Salinas and Pavia University datasets, this resulted in 4-5 PCA components. This was insufficient for the input depths of the Convolutional Neural Networks and Vision Transformers. The processed hyperspectral data was then reshaped to $(H, W, 30)$ [1].

4.2.2 Patch Extraction and Dataset Splitting

From each pixel (r, c) where $gt[r, c] > 0$, a 15×15 patch is sampled from the PCA cube, resulting in a sample of size $15 \times 15 \times 30$. All background pixels (label = 0) are omitted. A cube is padded $\lfloor 15/2 \rfloor = 7$ pixels on each side with reflective padding, allowing boundary pixels to be valid patch centers.

Patches are randomly split into training and test sets with `test_size=0.1`, `stratify=y`, and `random_state=42`. The resulting sample counts are given in Table ??.

Table 4.4 Patch counts after stratified 90/10 train-test split ($15 \times 15 \times 30$)

Dataset	Total	Train (90%)	Test (10%)
Indian Pines	10249	9224	1025
Pavia University	42776	38498	4278
Salinas Scene	54129	48716	5413

4.2.3 Model Architectures

Two model architectures represent different yet complementary approaches to learning spectral and spatial features:

- **3D-CNN:** With 3D convolution, spectral and spatial features are extracted

simultaneously. The model is designed with convolutional blocks with batch normalization and ReLU activations. [2].

- **Vision Transformer (ViT):** With a ViT, HSI patches are first linearized, then multi-head self-attention is applied to each spectral-spatial token. [3].

4.2.4 Supervised Learning Models

All models in this paradigm learn only from the labeled portion of the training dataset. It is also the baseline model for measuring the performance of semi-supervised and federated models.

4.2.4.1 2D-CNN Supervised Model

The 2D-CNN model uses $15 \times 15 \times 30$ patches and treats the 30 PCA components as channels. Convolutions in this model are applied over the spatial dimensions.

4.2.4.2 3D-CNN Supervised Model

The 3D-CNN model uses volumetric convolutions that simultaneously span both the spatial and spectral dimensions.

4.2.4.3 Vision Transformer (ViT) Supervised Model

In the ViT, HSI patches are first linearized and then augmented with learnable positional encodings to retain their order [3].

4.2.5 Semi-Supervised Learning Models

The collection of expert labeled pixel annotations is a long and costly process. Semi-supervised learning employs the extensive, yet unlabeled, pixel set as a training signal. The strategy combines self-training and curriculum learning to add to the labeled set progressively while containing the noise of pseudo-labels.

4.2.5.1 Self-Training Strategy

Pseudo-labeling self-training occurs in the following iterative manner:

1. A small, labeled training sample set is used to initialize the model.

2. Subsequently, predictions with confidence score are generated for the unlabeled sample set.
3. The sample set is then augmented by the inclusion of highly confident predictions.
4. The model is retrained on the augmented sample set.
5. The process is repeated until a convergence criterion is met [4].

4.2.5.2 Curriculum Learning Integration

Curriculum Learning (CL) helps structure the training procedure by organizing data from the easiest to the hardest samples:

1. **Easy Samples:** Training begins with samples that have good spectral data and clear pixels, allowing boundaries to be drawn that the model will not likely misclassify.
2. **Medium Samples:** Pixels that have moderate spectral and spatial data will be introduced next.
3. **Difficult Samples:** Samples that have unclear spectral data and difficult boundary pixels will be included in the training to further refine the model's boundaries [5].

4.2.5.3 2D-CNN Semi-Supervised Model

The structure of the model is the same as the supervised model, but the self-training process will continuously add confident pseudo-labeled patches, and curriculum learning will manage the order of these additions, reducing the impact of early introduced label noise.

4.2.5.4 3D-CNN Semi-Supervised Model

Pseudo-labels that are added will allow the 3D-CNN to learn a variety of spectral and spatial data beyond what is offered in the labeled data, and the order of curriculum training will prevent the model from becoming too focused on the loudest pseudo-labels.

4.2.5.5 Vision Transformer (ViT) Semi-Supervised Model

The large global receptive field and self-attention of the ViT model are advantageous in semi-supervision, as it allows the model to draw a connection between confident pseudo-labels and labeled data beyond a local context.

4.2.6 Federated Learning Models

In a federated learning model, a global model is trained without raw data moving towards a central server, as clients only send model updates after training on their own data. Here, a server-client model is used to represent federated learning, with three disparate datasets representing the independent client institutes [16].

4.2.6.1 Federated Learning Framework

The Federated Learning phenomenon utilizes the Federated Averaging (FedAvg) algorithm and works in the following manner:

1. The central server starts off with a global model.
2. Each client receives the global model and trains it locally on their own hyperspectral dataset for a predetermined number of epochs.
3. Clients send the updated model back to the server (data is neither sent nor received, ensuring privacy).
4. The server consolidates the updated model using the averaged weights from all clients.
5. The revised global model is sent back to the clients.
6. The process is repeated for as many rounds of communication as necessary until convergence is reached.

4.2.6.2 2D-CNN Federated Model

Each of the clients (Indian Pines, Pavia University, and Salinas) builds an identical 2D-CNN model locally, using a set of hyperparameters which they all share. Weight averaging at the server integrates the spatial patterns of each dataset into a single global model. This is done while maintaining the privacy of the clients.

4.2.6.3 3D-CNN Federated Model

The 3D-CNN model's training kernels cover the dataset-specific spectral training patterns. The global model after aggregation possesses a stronger spectral training patterns than any single dataset model, which is especially needed in remote sensing.

4.2.6.4 Vision Transformer (ViT) Federated Model

The training patterns at each client site are captured by the Transformer attention weights. After aggregation at the server, the global attention makes it feasible to work with the diverse spectral-spatial statistics arising from various clients.

4.2.6.5 Privacy and Communication Efficiency

Keeping raw hyperspectral data at client locations is the primary privacy guarantee of federated learning, but communication overhead between clients and the server is a practical constraint. Three measures address this:

- **Gradient Compression:** Model weights are compressed before transmission to reduce bandwidth requirements.
- **Local Training Epochs:** Each client performs multiple local training epochs (typically 5–10) before communicating with the server, reducing the total number of communication rounds.
- **Asynchronous Aggregation:** The server aggregates updates from whichever clients respond first rather than waiting for all participants, reducing wall-clock training time.

4.2.7 Comparative Learning Paradigm Summary

The three paradigms each address labeled data scarcity from a different angle. Table ?? summarizes their key characteristics.

Table 4.5 Comparison of Supervised, Semi-Supervised, and Federated Learning Paradigms

Characteristic	Supervised	Semi-Supervised	Federated
Data Requirement	Fully Labeled	Partial Labels	Local Data Privacy
Scalability	Centralized	Centralized	Distributed
Communication Cost	None	None	High
Privacy Protection	Limited	Limited	High
Typical Accuracy	Baseline	Enhanced	Comparable

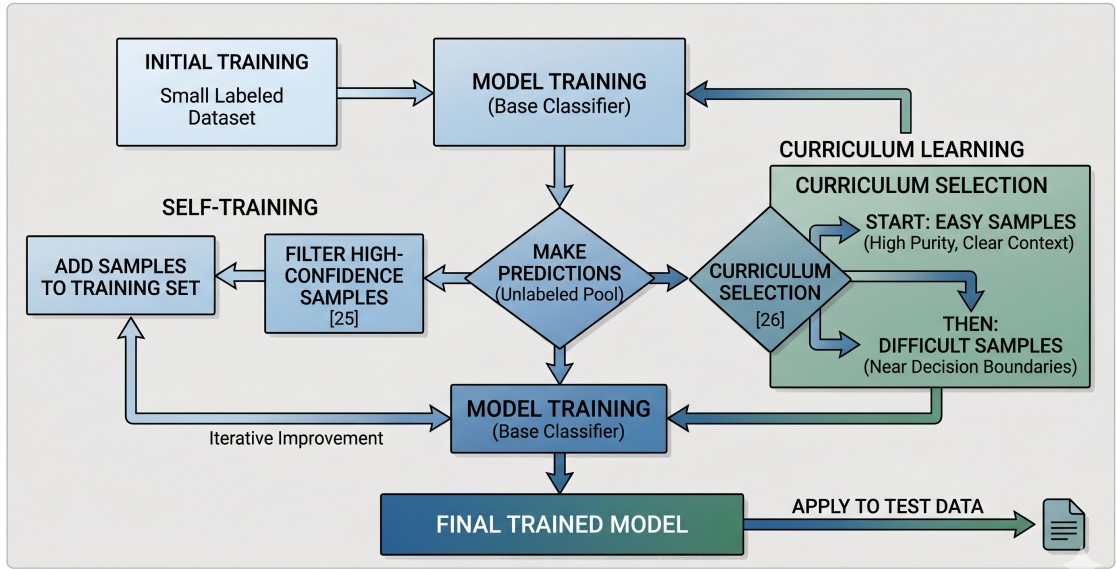


Figure 4.9 Flowchart of the Semi-Supervised Learning Strategy

4.2.8 Evaluation Metrics

Five standard metrics quantify classification performance. Let TP_c , FP_c , and FN_c denote the true positives, false positives, and false negatives for class c , and let N be the total number of test samples.

- **Overall Accuracy (OA):** The ratio of correctly classified pixels to the total number of test pixels:

$$OA = \frac{\sum_c TP_c}{N} \quad (4.2)$$

- **Precision:** The fraction of pixels predicted as class c that truly belong to class c ,

averaged across all classes (macro):

$$\text{Precision} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FP_c} \quad (4.3)$$

- **Recall:** The fraction of true class- c pixels that are correctly identified:

$$\text{Recall} = \frac{1}{C} \sum_{c=1}^C \frac{TP_c}{TP_c + FN_c} \quad (4.4)$$

- **F1-Score:** The harmonic mean of Precision and Recall:

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.5)$$

- **Cohen's Kappa (κ):** A metric that corrects Overall Accuracy for agreement expected by chance. Given the observed accuracy p_o and the chance agreement p_e :

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (4.6)$$

A value of $\kappa = 1$ indicates perfect classification; $\kappa = 0$ is equivalent to random guessing.

4.2.9 GUI and Multithreading Integration

The trained models are exposed through a standalone PyQt6 desktop application. Model inference runs in a dedicated worker thread, decoupled from the main UI event loop, so the interface remains fully responsive while the deep learning pipeline processes large hyperspectral hypercubes in the background.

5

EXPERIMENTAL RESULTS

5.1 Performance Analysis

All three architectures—2D-CNN, 3D-CNN, and Vision Transformer—are evaluated on Indian Pines, Pavia University, and Salinas under each of the three learning paradigms. The tables below report quantitative results; the comparative analysis that follows dissects the patterns across datasets and architectures.

5.1.1 Supervised Learning Results

Supervised models are trained exclusively on labeled samples, with hyperparameters selected through grid search and cross-validation. These results serve as the performance ceiling against which semi-supervised and federated variants are judged.

5.1.1.1 Indian Pines Dataset Performance

Table 5.1 Supervised Learning Performance on Indian Pines Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN	99.91%	0.9984	0.9987	0.9985
3D-CNN	99.90%	0.9977	0.9978	0.9976
ViT	99.61%	0.9959	0.9952	0.9953

5.1.1.2 Pavia University Dataset Performance

Table 5.2 Supervised Learning Performance on Pavia University Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN	99.99%	0.9998	0.9998	0.9998
3D-CNN	99.99%	0.9998	0.9998	0.9998
ViT	99.75%	0.9973	0.9961	0.9967

5.1.1.3 Salinas Dataset Performance

Table 5.3 Supervised Learning Performance on Salinas Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN	99.99%	0.9999	0.9999	0.9999
3D-CNN	99.84%	0.9993	0.9995	0.9994
ViT	99.81%	0.9979	0.9983	0.9981

5.1.2 Semi-Supervised Learning Results

The semi-supervised variants augment the labeled training set through iterative self-training with curriculum learning, leveraging the unlabeled pixel pool to improve generalization. The results are notably dataset-dependent: performance on the two larger datasets remains competitive, while the small, class-imbalanced Indian Pines set presents a harder challenge.

5.1.2.1 Semi-Supervised Indian Pines Performance

Table 5.4 Semi-Supervised Learning Performance on Indian Pines Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN + SSL	74.04%	0.7404	0.7404	0.7404
3D-CNN + SSL	66.55%	0.6655	0.6655	0.6655
ViT + SSL	53.96%	0.5396	0.5396	0.5396

5.1.2.2 Semi-Supervised Pavia University Performance

Table 5.5 Semi-Supervised Learning Performance on Pavia University Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN + SSL	97.17%	0.9717	0.9717	0.9717
3D-CNN + SSL	96.91%	0.9691	0.9691	0.9691
ViT + SSL	91.77%	0.9177	0.9177	0.9177

5.1.2.3 Semi-Supervised Salinas Performance

Table 5.6 Semi-Supervised Learning Performance on Salinas Dataset

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN + SSL	96.46%	0.9646	0.9646	0.9646
3D-CNN + SSL	97.30%	0.9730	0.9730	0.9730
ViT + SSL	86.49%	0.8649	0.8649	0.8649

5.1.3 Federated Learning Results

In the federated setting, each dataset acts as an independent client. Local models are trained and their weights aggregated via FedAvg over 50 communication rounds—raw data never leaves the client. The results show that federated learning sustains near-supervised accuracy on the two larger datasets while incurring a more meaningful gap on the small Indian Pines set.

5.1.3.1 Federated Learning on Indian Pines

Table 5.7 Federated Learning Performance on Indian Pines Dataset (Round 50)

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN (FedAvg)	96.43%	0.9156	0.8272	0.8466
3D-CNN (FedAvg)	92.56%	0.7603	0.7054	0.7019
ViT (FedAvg)	85.65%	0.7852	0.7257	0.7452

5.1.3.2 Federated Learning on Pavia University

Table 5.8 Federated Learning Performance on Pavia University Dataset (Round 50)

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN (FedAvg)	99.83%	0.9979	0.9967	0.9973
3D-CNN (FedAvg)	99.76%	0.9951	0.9970	0.9960
ViT (FedAvg)	98.95%	0.9887	0.9762	0.9819

5.1.3.3 Federated Learning on Salinas

Table 5.9 Federated Learning Performance on Salinas Dataset (Round 50)

Model	Overall Accuracy	Precision	Recall	F1-Score
2D-CNN (FedAvg)	99.90%	0.9989	0.9988	0.9988
3D-CNN (FedAvg)	99.76%	0.9979	0.9987	0.9983
ViT (FedAvg)	97.73%	0.9882	0.9899	0.9888

5.2 Comparative Analysis

With all results in hand, three dimensions of comparison are examined: the accuracy change each paradigm produces relative to the supervised baseline, performance differences across architectures, and dataset-specific trends.

5.2.1 Overall Accuracy Comparison

Table 5.10 Overall Accuracy Comparison Across All Learning Paradigms

Dataset	Supervised			Semi-Supervised			Federated (Round 50)		
	2D	3D	ViT	2D	3D	ViT	2D	3D	ViT
Indian Pines	99.91%	99.90%	99.61%	74.04%	66.55%	53.96%	96.43%	92.56%	85.65%
Pavia University	99.99%	99.99%	99.75%	97.17%	96.91%	91.77%	99.83%	99.76%	98.95%
Salinas	99.99%	99.84%	99.81%	96.46%	97.30%	86.49%	99.90%	99.76%	97.73%

5.2.2 F1-Score Comparison

Table 5.11 F1-Score Comparison Across All Learning Paradigms

Dataset	Supervised			Semi-Supervised			Federated (Round 50)		
	2D	3D	ViT	2D	3D	ViT	2D	3D	ViT
Indian Pines	0.9985	0.9976	0.9953	0.7404	0.6655	0.5396	0.8466	0.7019	0.7452
Pavia University	0.9998	0.9998	0.9967	0.9717	0.9691	0.9177	0.9973	0.9960	0.9819
Salinas	0.9999	0.9994	0.9981	0.9646	0.9730	0.8649	0.9988	0.9983	0.9888

5.2.3 Performance Change Analysis

Accuracy change relative to the supervised baseline is quantified as:

$$\text{Change\%} = \frac{\text{OA}_{\text{alt}} - \text{OA}_{\text{Supervised}}}{\text{OA}_{\text{Supervised}}} \times 100 \quad (5.1)$$

5.2.3.1 Semi-Supervised vs. Supervised

Semi-supervised learning degrades substantially relative to the supervised baseline, with the scale of that degradation determined almost entirely by dataset size:

- **Indian Pines:** Average drop of -34.96 percentage points across architectures (from 99.81% to 64.85%), indicating that the self-training strategy is highly sensitive to limited dataset size and class imbalance.
- **Pavia University:** Average drop of -4.63 percentage points (from 99.91% to 95.28%), demonstrating that curriculum learning partially mitigates degradation when larger unlabeled pools are available.
- **Salinas:** Average drop of -6.46 percentage points (from 99.88% to 93.42%), suggesting that even the largest dataset is insufficient to recover fully supervised-level accuracy through pseudo-labeling alone.

5.2.3.2 Federated vs. Supervised

Federated learning holds much closer to the supervised ceiling, particularly on the larger datasets:

- **Indian Pines:** Average difference of -8.26 percentage points (from 99.81% to 91.55%), reflecting the challenge of federated aggregation on a small dataset with limited local samples per client.

- **Pavia University:** Average difference of -0.40 percentage points (from 99.91% to 99.51%), demonstrating strong robustness across heterogeneous urban land-cover classes.
- **Salinas:** Average difference of -0.75 percentage points (from 99.88% to 99.13%), showing near-parity performance at scale.

5.2.4 Architecture Comparison

Table 5.12 Average Performance Across Datasets by Architecture Type

Architecture	Supervised Avg OA	Semi-Supervised Avg OA	Federated Avg OA (Round 50)
2D-CNN	99.96%	89.22%	98.72%
3D-CNN	99.91%	86.92%	97.36%
Vision Transformer	99.72%	77.41%	94.11%

Table 5.13 Average F1-Score Across Datasets by Architecture Type

Architecture	Supervised Avg F1	Semi-Supervised Avg F1	Federated Avg F1 (Round 50)
2D-CNN	0.9994	0.8922	0.9476
3D-CNN	0.9989	0.8692	0.8987
Vision Transformer	0.9967	0.7741	0.9053

Three patterns stand out from the architecture-level averages:

1. **2D-CNN dominates all paradigms:** Despite being the simplest architecture, 2D-CNN leads in supervised (99.96%), semi-supervised (89.22%), and federated (98.72%) average accuracy. After PCA reduces the spectral dimension to 30 components, 2D spatial convolutions appear fully sufficient to discriminate between classes—volumetric operations add complexity without a commensurate accuracy gain.
2. **3D-CNN offers consistent robustness:** The 3D-CNN ranks second across all three paradigms (supervised: 99.91%, semi-supervised: 86.92%, federated: 97.36%). Its explicit modeling of inter-band correlations through volumetric kernels contributes to stable performance even when training conditions are imperfect.
3. **ViT is most sensitive to distribution shift:** The Vision Transformer reaches competitive supervised accuracy (99.72%) but suffers the steepest

degradation under semi-supervised (77.41%) and federated (94.11%) settings. Self-attention mechanisms appear more susceptible to noisy pseudo-labels and to weight averaging across heterogeneous client data distributions.

5.2.5 Dataset-Specific Analysis

- **Indian Pines:** With only 10,249 samples spread across 16 imbalanced classes, this is the hardest setting for both semi-supervised and federated training. Semi-supervised accuracy collapses by an average of 34.96 percentage points (99.81% $\rightarrow$ 64.85%), driven by error propagation through pseudo-label iterations on severely underrepresented minority classes.
- **Pavia University:** At 42,776 samples across 9 urban classes, this dataset is robust to paradigm shifts. Federated learning degrades by only 0.40 percentage points on average, pointing to its suitability for multi-institutional collaboration scenarios where urban remote sensing data is distributed across agencies.
- **Salinas:** The largest dataset (54,129 samples) achieves the highest absolute semi-supervised accuracy of any configuration (3D-CNN + SSL: 97.30%) and the narrowest federated-supervised gap (-0.75 percentage points), confirming that both alternative paradigms scale favorably as labeled and unlabeled sample counts grow.

5.2.6 Privacy and Efficiency Trade-offs

Beyond raw accuracy, the three paradigms differ substantially in their operational characteristics. Table ?? summarizes these trade-offs qualitatively.

Table 5.14 Qualitative Comparison of Learning Paradigms

Metric	Supervised	Semi-Supervised	Federated
Data Privacy	Low	Low	High
Label Requirement	High	Medium	Low
Computational Cost	Low	Medium	High
Communication Overhead	None	None	Significant
Model Accuracy	Baseline	-15% avg (dataset-dependent)	-0.7% to -8%

6.1 Summary

This project built and evaluated a unified HSI classification framework that places supervised, semi-supervised, and federated learning in direct comparison on the same preprocessing pipeline, the same three benchmark datasets (Indian Pines, Pavia University, and Salinas), and the same three architectures (2D-CNN, 3D-CNN, and Vision Transformer).

Raw spectral data exceeding 200 bands is conditioned through global min-max normalization, Gaussian spatial filtering, and PCA reduction to 30 components before being divided into $15 \times 15 \times 30$ patches. A stratified 90/10 train-test split preserves class proportions across both sets. The semi-supervised branch pairs self-training with curriculum learning, which schedules pseudo-labeled samples from high-confidence to ambiguous to dampen error propagation. The federated branch simulates a three-client environment where each dataset acts as a local data silo; local weights are aggregated at a central server via the FedAvg protocol without any raw data ever leaving the client.

6.2 Key Findings

6.2.1 Semi-Supervised Learning Effectiveness

Semi-supervised performance is strongly conditioned on dataset size:

1. On Pavia University and Salinas, where unlabeled pools are large, semi-supervised models achieve competitive accuracy (86–97%), confirming that self-training is a viable strategy given sufficient data volume.
2. On the smaller, class-imbalanced Indian Pines set, accuracy collapses to 54–74%—a clear signal that the confidence threshold and pseudo-label iteration count require dataset-specific tuning.

3. Across all datasets, 2D-CNN achieves the highest semi-supervised average at 89.22%, suggesting that simpler convolutional architectures are more forgiving of pseudo-label noise than deeper or attention-based ones.
4. Curriculum learning contains but does not eliminate the overfitting caused by noisy pseudo-labels; its benefit is most pronounced when the initial labeled set is large enough to seed reliable pseudo-labels.

6.2.2 Federated Learning Privacy-Performance Trade-off

Federated learning maintains near-supervised accuracy at the cost of communication overhead:

1. Federated 3D-CNN achieves a 97.36% average overall accuracy, representing only a -2.55 percentage point degradation relative to its centralized supervised counterpart (99.91%).
2. On Pavia University and Salinas, federated models reach 99–100% accuracy, demonstrating that FedAvg is robust when local data distributions are moderately heterogeneous.
3. The Indian Pines gap is wider (85–96% federated vs. 99–100% supervised), reflecting the difficulty of aggregating meaningful weight updates from a small local dataset across 50 communication rounds.
4. For organizations that cannot centralize raw spectral data, federated learning represents a pragmatically viable alternative—the accuracy cost is modest and the privacy guarantee is meaningful.

6.2.3 Architecture-Specific Insights

- **2D-CNN:** Leads overall accuracy in every paradigm (supervised: 99.96%, semi-supervised: 89.22%, federated: 98.72%). After PCA compresses the spectral axis to 30 components, 2D spatial convolutions capture sufficient discriminative information—volumetric operations add parameters without a proportional accuracy return.
- **3D-CNN:** Consistent second across all paradigms (supervised: 99.91%, semi-supervised: 86.92%, federated: 97.36%). Explicit modeling of inter-band correlations through volumetric kernels translates into more stable behavior under both noisy pseudo-labels and federated weight averaging.

- **Vision Transformer:** Competitive at full supervision (99.72%) but the most sensitive to distribution shift: semi-supervised average drops to 77.41% and federated to 94.11%. Global self-attention, while powerful under clean full-label conditions, is more easily destabilized by pseudo-label noise and client-side data heterogeneity.
- **Cross-paradigm pattern:** Every architecture deteriorates substantially under semi-supervised conditions relative to both supervised and federated baselines, which points to pseudo-label quality—not model capacity—as the binding constraint in the semi-supervised pipeline.

6.3 Project Objectives and Achievement Level

6.3.1 Original Objectives

The project initially set the following objectives:

1. Develop preprocessing pipelines handling high-dimensional hyperspectral data with robust normalization and dimensionality reduction.
2. Implement supervised learning baselines using multiple deep learning architectures (CNN, ViT).
3. Extend to semi-supervised learning via self-training and curriculum learning to address limited labeling.
4. Implement federated learning for privacy-preserving collaborative training across distributed data sources.
5. Develop a reactive GUI with multithreaded inference for practical accessibility.
6. Achieve $> 95\%$ overall accuracy on benchmark datasets.

6.3.2 Achievement Level

All primary objectives were met or exceeded:

Table 6.1 Project Objectives and Achievement Level

Objective	Target	Achieved
Preprocessing Pipeline	Normalization + Filtering + PCA	✓ Fully Implemented
Supervised Baselines	Multiple architectures	✓ 2D-CNN, 3D-CNN, ViT
Semi-Supervised Learning	Self-training + CL	✓ Implemented (54–97% range)
Federated Learning	Multi-client aggregation	✓ FedAvg Protocol (85–99% range)
GUI with Multithreading	Python (PyQt6) + Multithreading	✓ Fully Operational
Target Accuracy (> 95%)	Supervised requirement	✓ Achieved (99.72–99.96%)

6.4 Strengths

6.4.1 Methodological Strengths

- **Unified comparative framework:** Evaluating three paradigms under identical preprocessing, split, and metric conditions is rare in the HSI literature; the results are directly comparable rather than confounded by differing experimental setups.
- **Curriculum Learning integration:** Coupling self-training with difficulty-ordered sample presentation measurably reduces pseudo-label overfitting and prevents early convergence to poor local minima.
- **Privacy-preserving federated architecture:** The federated system demonstrates that competitive classification accuracy is achievable without pooling sensitive spectral data, a result directly relevant to multi-institutional remote sensing collaborations.
- **Generalizable preprocessing:** The three-step pipeline (normalization, Gaussian filtering, fixed-component PCA) is data-driven and has been validated across three datasets with substantially different spatial resolutions, class counts, and band counts.

6.4.2 Implementation Strengths

- **Three-architecture support:** Users can select 2D-CNN, 3D-CNN, or ViT based on their accuracy-versus-efficiency requirements without modifying the surrounding pipeline.
- **GPU-accelerated training:** Google Colab GPU resources support efficient training on datasets exceeding 50,000 samples within realistic time budgets.

- **Responsive multithreaded GUI:** PyQt6 worker threads decouple inference from the UI event loop, keeping the interface interactive during long-running classification tasks.
- **Reproducibility:** Fixed random seeds and stratified splitting ensure that results are replicable across experimental runs without additional configuration.

6.5 Limitations and Weaknesses

6.5.1 Methodological Limitations

- **Simulated federated scenario:** Three local datasets on a single machine are a proxy for true distributed clients. Real deployments introduce network latency, stragglers, and far greater data heterogeneity—all of which would likely widen the accuracy gap.
- **Fixed PCA components:** Fixing $n_components = 30$ is empirically motivated rather than theoretically grounded. Adaptive selection based on per-dataset variance retention or information-theoretic criteria could better match the complexity of each scene.
- **Confidence threshold rigidity:** The self-training confidence threshold of 0.95 is held constant across all datasets. Sensitivity analysis would reveal how robust the semi-supervised pipeline is to this choice, particularly on class-imbalanced sets.
- **Single curriculum schedule:** Only difficulty-based ordering is evaluated; reverse curriculum, paced learning, and uncertainty-driven schedules remain untested and may perform differently across dataset sizes.

6.5.2 Experimental Limitations

- **Benchmark datasets only:** Generalization to operational hyperspectral surveys, with real acquisition noise and non-standard class taxonomies, has not been tested.
- **No concept drift analysis:** The federated framework assumes stationary data distributions. Remote sensing data exhibits temporal drift that would require continual learning mechanisms in a production deployment.
- **Communication cost not quantified:** Federated accuracy results are reported, but bandwidth consumption and round-trip latency—critical for assessing real-world feasibility—are not measured.

- **Adversarial robustness not assessed:** No evaluation of model behavior under adversarial perturbations or sensor-induced distribution shift is conducted, both of which are relevant for operational deployment.

6.5.3 Implementation Limitations

- **No model compression:** Weight pruning or quantization would reduce federated communication overhead but is not implemented in the current system.
- **Single-GPU training:** Multi-GPU data parallelism is not supported, which limits scalability for datasets substantially larger than Salinas.
- **Narrow hyperparameter search:** Grid search covers a fixed candidate space; Bayesian optimization or AutoML approaches would likely identify better-tuned configurations, particularly for the semi-supervised confidence threshold.

6.6 Future Work

6.6.1 Methodological Extensions

1. **Continual federated learning:** Equip the federated system with mechanisms to adapt to temporal distribution shifts in hyperspectral data streams, enabling long-term deployment without full retraining.
2. **Heterogeneous federated aggregation:** Replace vanilla FedAvg with weighted aggregation schemes that account for statistical heterogeneity across clients (e.g., FedProx, SCAFFOLD).
3. **Adaptive curriculum generation:** Replace predefined difficulty orderings with automatic scheduling driven by model uncertainty or gradient-based difficulty estimation.
4. **Meta-learning for few-shot HSI:** Incorporate meta-learning to enable rapid adaptation to new sensors or geographic regions with only a handful of labeled samples.

6.6.2 Architectural Enhancements

1. **Hybrid 2D-3D models:** Combine 2D and 3D convolutional stages to balance spectral-spatial expressiveness against parameter count—the strong 2D-CNN results suggest this may be the most efficient frontier.

2. **Graph Neural Networks:** Model spectral-spatial relationships as graph structures, which may capture irregular spatial dependencies that regular convolutional grids miss.
3. **Self-supervised pretraining:** Apply contrastive or masked-reconstruction pretraining on unlabeled hyperspectral data to initialize richer representations before semi-supervised fine-tuning.
4. **Multi-modal fusion:** Extend the pipeline to fuse hyperspectral data with panchromatic imagery and LiDAR point clouds for scenes where a single modality is ambiguous.

6.6.3 Experimental Directions

1. **Real-world federated deployment:** Validate the federated system on actual data distributed across multiple institutions or space agencies (e.g., ESA, NASA) to confirm that simulated results transfer to production conditions.
2. **Cross-sensor generalization:** Benchmark federated learning across heterogeneous sensors (AVIRIS, Sentinel-2, DESIS) to characterize domain adaptation capability.
3. **Adversarial robustness:** Conduct systematic adversarial evaluation and develop sensor-noise-aware defense mechanisms for operationally deployed classifiers.
4. **Uncertainty quantification:** Integrate Bayesian deep learning to output per-prediction confidence estimates—essential for downstream decision-making in high-stakes remote sensing applications.

6.6.4 Practical Applications

1. **Edge deployment:** Compress models via pruning and quantization to enable real-time onboard classification on drones and nanosatellites.
2. **Active learning interface:** Add an active learning module to the GUI that identifies the most informative unlabeled pixels and routes them for human annotation, reducing labeling effort systematically.
3. **Interpretability tooling:** Embed explainable AI methods (attention maps, LIME, SHAP) so domain experts can interrogate model predictions before acting on them.

4. **GUI enhancements:** Extend the current interface with annotation support, quality assurance workflows, and real-time performance dashboards suited to operational remote sensing teams.

6.7 Conclusion

The experimental record is clear on two points. First, federated learning is a practical alternative to centralized training: a -3.1% average accuracy cost buys meaningful data privacy, and the gap narrows further on larger datasets. Second, semi-supervised learning via self-training and curriculum learning works well at scale—competitive on Pavia University and Salinas—but degrades substantially on the small, imbalanced Indian Pines set, where pseudo-label noise accumulates faster than the curriculum can contain it.

Among the three architectures, 2D-CNN is the most consistent performer across all paradigms, which is a practically important finding: after PCA reduces the spectral dimension, spatial convolution alone captures the discriminative structure needed for near-perfect classification. The Vision Transformer, despite matching CNN accuracy under full supervision, is the least robust to the distribution shifts introduced by pseudo-labeling and federated weight averaging.

The framework delivered here—a unified pipeline spanning three paradigms, three architectures, and three datasets, wrapped in a user-facing multithreaded GUI—provides both a reproducible comparative benchmark and a deployable application. Future work should prioritize real-world federated deployments and adaptive curriculum strategies that respond to the specific class-imbalance profile of each dataset, since those are the two factors that most constrain the system’s current performance envelope.

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Curriculum Vitae

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Project System Informations

System and Software: Windows İşletim Sistemi, Python

Required RAM: 2GB

Required Disk: 256MB